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Onbepaalde vormen en
de regel van L'Hôpital

Regel van L'Hôpital (John Bernoulli)

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = ? \quad 1$$

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = \frac{0}{0} \quad ?$$

$$\frac{\cos(0)}{1} = 1$$

Stelling: De eerste vorm van de regel van l'Hôpital

Beschouw twee functies f en g .

Stel dat $f(a)=g(a)=0$ en dat $f'(a)$ en $g'(a)$ bestaan met $g'(a) \neq 0$, dan geldt:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}$$

Bewijs **NIET**

Voorbeeld $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} = ?$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} &= \frac{\frac{0}{0}}{1} = \frac{\frac{1}{2\sqrt{1+0}}}{1} \\ &= \frac{1}{2} \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - x/2}{x^2} = ? \quad \left(\frac{0}{0}\right)$$

$$\frac{\frac{1}{2\sqrt{1+0}} - \frac{1}{2}}{2 \times 0} = \frac{\frac{0}{0}}{0} = \frac{0}{0}$$

$$\frac{1}{2} (1+x)^{-1/2}$$

Stelling: De sterke vorm van de regel van l'Hôpital

Beschouw twee functies f en g.

Stel dat $f(a)=g(a)=0$ en dat f en g differentieerbaar zijn op een open interval I rond a en dat $g'(x) \neq 0$ op I als $x \neq a$, dan geldt:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Indien de limiet in het rechterlid bestaat.

$$\begin{aligned} & \Rightarrow \lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{1+x}} - \frac{1}{2}}{2x} \\ & = \lim_{x \rightarrow 0} \frac{-\frac{1}{4} (1+x)^{-3/2}}{2} \\ & = \frac{-\frac{1}{4}}{2} = -\frac{1}{8} \end{aligned}$$

Bewijs mbv de middelwaardestelling van Cauchy: niet

Werkwijze: f en g blijven afleiden in a tot één van beide verschillend is van 0

$$\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x + x^2} = ? \quad \text{! Op tijd stoppen met afleiden !}$$

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$$= \lim_{x \rightarrow 0} \frac{\sin(x)}{1 + 2x} = \frac{0}{1} = 0$$

$$? \quad \cancel{\lim_{x \rightarrow 0} \frac{\cos(x)}{2}} = \frac{1}{2} \quad *$$

Toepassingen

1. linker en rechterlimieten

$$\lim_{x \rightarrow 0+} \frac{\sin(x)}{x^2}$$

$$\lim_{x \rightarrow 0-} \frac{\sin(x)}{x^2}$$

2. onbepaalde vormen

$$\frac{\frac{1}{0}}{\frac{1}{0}} = \frac{0}{0}$$



$$\frac{\infty}{\infty}$$

→

$$\lim_{x \rightarrow \infty} \frac{\ln(x)}{2\sqrt{x}}$$

$$\infty \cdot 0$$

→

$$\lim_{x \rightarrow +\infty} x \sin\left(\frac{1}{x}\right)$$

$$\infty - \infty$$

→

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin(x)} - \frac{1}{x} \right)$$

3. onbepaalde machten

$$1^\infty$$

→

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x$$

= Definitie van getal e

$$0^0$$

→

$$\lim_{x \rightarrow 0+} (x^x)$$

$$\infty^0$$

→

$$\lim_{x \rightarrow \infty} \left(x^{\frac{1}{x}} \right)$$

Gebruik: $\lim_{x \rightarrow a} \ln(f(x)) = L \Rightarrow \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} e^{\ln(f(x))} = e^{\lim_{x \rightarrow a} \ln(f(x))} = e^L$

1. linker en rechterlimieten

$$\lim_{x \rightarrow 0^+} \frac{\sin(x)}{x^2}$$

$$\stackrel{\left(\frac{0}{0}\right)}{=} \lim_{x \rightarrow 0^+} \frac{\cos(x)}{2x} = \lim_{x \rightarrow 0^+} \frac{1}{2x} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{\sin(x)}{x^2}$$

$$\begin{aligned} &\stackrel{\left(\frac{0}{0}\right)}{=} \lim_{x \rightarrow 0^-} \frac{\cos(x)}{2x} \\ &= \lim_{x \rightarrow 0^-} \frac{1}{2x} \\ &= -\infty \end{aligned}$$

$$\lim_{x \rightarrow \infty} \frac{\ln(x)}{2\sqrt{x}} \stackrel{\left(\frac{\infty}{\infty}\right)}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{\sqrt{x}}}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0$$

2. onbepaalde vormen

$$\frac{\infty}{\infty} \longrightarrow \lim_{x \rightarrow \infty} \frac{\ln(x)}{2\sqrt{x}}$$

$$\infty \cdot 0 \longrightarrow \lim_{x \rightarrow +\infty} x \sin\left(\frac{1}{x}\right)$$

$$\infty - \infty \longrightarrow \lim_{x \rightarrow 0} \left(\frac{1}{\sin(x)} - \frac{1}{x} \right)$$

$$\longrightarrow \lim_{x \rightarrow +\infty} \cancel{x} \frac{\sin\left(\frac{1}{x}\right)}{\frac{1}{x}} \stackrel{\left(\frac{0}{0}\right)}{=} \lim_{x \rightarrow +\infty} \frac{\cos\left(\frac{1}{x}\right) \left(-\frac{1}{x^2}\right)}{\left(-\frac{1}{x^2}\right)}$$

$$= \lim_{x \rightarrow +\infty} \cos\left(\frac{1}{x}\right) = 1$$

$$= \lim_{x \rightarrow 0} \frac{x - \sin(x)}{x \sin(x)}$$

$$\stackrel{\left(\frac{0}{0}\right)}{=} \lim_{x \rightarrow 0} \frac{1 - \cos(x)}{\sin(x) + x \cos(x)} \stackrel{\left(\frac{0}{0}\right)}{=} \lim_{x \rightarrow 0} \frac{\sin(x)}{\cos(x) + \cos(x) - x \sin(x)} = \frac{0}{1+1+0} = \frac{0}{2} = 0$$

3. onbepaalde machten

$$\begin{aligned} 1^\infty \\ 0^0 \\ \infty^0 \end{aligned}$$

$$\begin{aligned} &\longrightarrow \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = \text{Definitie van getal } e \\ &\longrightarrow \lim_{x \rightarrow 0^+} (x^x) = 1 \\ &\longrightarrow \lim_{x \rightarrow \infty} \left(x^{\frac{1}{x}}\right) = 1 \end{aligned}$$

$$\textcircled{1} \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = e^1 = e$$

NB: $\lim_{x \rightarrow \infty} x \ln\left(1 + \frac{1}{x}\right)$

$$= \lim_{x \rightarrow \infty} \frac{\ln\left(1 + \frac{1}{x}\right)}{1/x}$$

$$\stackrel{(0/0)}{=} \lim_{x \rightarrow \infty} \frac{\left(\frac{1}{1 + \frac{1}{x}}\right) \left(-\frac{1}{x^2}\right)}{-\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{1}{x}} = 1$$

$$\lim_{x \rightarrow a} f(x) \xrightarrow{\ln} \lim_{x \rightarrow a} \ln(f(x)) = L$$

$$\downarrow e^L$$

$$\xleftarrow{\exp}$$

$$\textcircled{2} \lim_{x \rightarrow 0^+} (x^x) \rightarrow \text{NB: } \lim_{x \rightarrow 0^+} (x \ln x) = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} \stackrel{(0/0)}{=} \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} (-x) = 0$$

$$e^0 = 1$$

$$\textcircled{3} \lim_{x \rightarrow \infty} (x^{1/x}) \rightarrow \text{NB: } \lim_{x \rightarrow \infty} \frac{1}{x} \ln x \stackrel{(0/0)}{=} \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\parallel e^0 = 1$$